

A PROJECT REPORT ON



COMPUTER NETWORKS

A HYPOTHETICAL CAMPUS NETWORK

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Computer Networks Mini Project Report

The proposed network setup for the hypothetical campus is divided into three main sections: Area 0, Area 1, and Area 2. Each area serves specific functions and is designed to ensure efficient communication across the campus.

1. Area 0: Central Backbone

Acts as the central backbone of the network, handling inter-area communication and serving as the connection point to the ISP (Internet Service Provider).

DNS Server

This DNS server handles all domain name resolution requests within Area 0 and forwards unresolved requests to the ISP.

Connection to ISP

All outbound requests from the campus network are routed through Area 0 to the ISP network, which includes a Web server and a DNS server.

2. Area 1: Hostels

Area 1 consists of two primary subnets, one for the boys' hostels and the other for the girls' hostels.

DHCP Servers

Both subnets are equipped with DHCP servers to dynamically allocate IP addresses to devices within the hostels.

Web Server

A web server is present in Area 1 to handle local web traffic.

DNS Resolution

Since Area 1 lacks its own DNS server, all DNS requests from this area are forwarded to the DNS server in Area 0 for name resolution.

3. Area 2: Departments and Labs

Area 2 has multiple subnets, each representing different departments, including Labs, Computer, Civil, Architecture, and Mechanical departments.

VLAN Configuration

The lab's subnet within Area 2 is divided into three VLANs: VLAN 10 (Computer Labs), VLAN 20 (Civil Labs), and VLAN 30 (Mechanical Labs). These VLANs are represented by vertical yellow rectangles in the network diagram.

Web Server

The Computer department's subnet includes a dedicated web server for departmental use.

DNS Server

Area 2 has its own DNS server to handle name resolution requests within the area, reducing the need for cross-area communication for DNS queries.

4. IP Addressing and Routing

IP Range

The entire campus network is assigned the IP range 7.7.0.0/22, which supports a maximum of 1022 hosts. However, due to subnetting and the specific network architecture, the actual number of hosts accommodated will be less than this maximum.

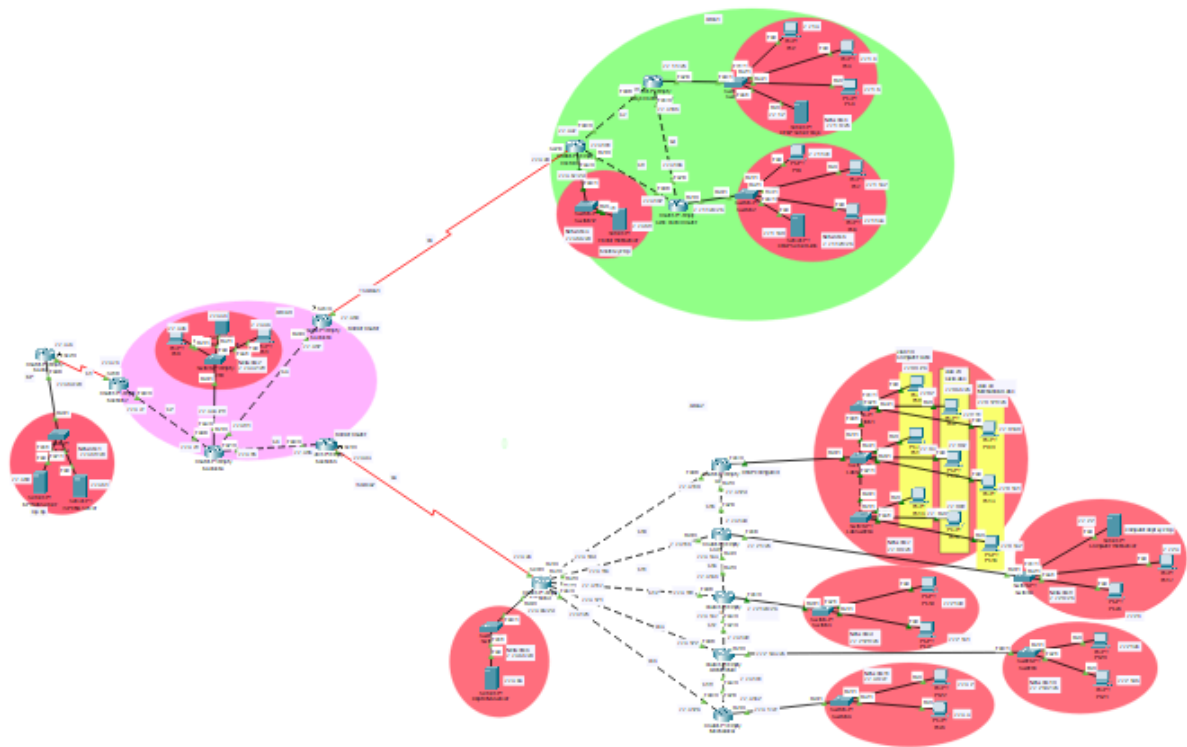
Border Routers

Each area is equipped with a dedicated border router to manage the ISP connection and inter-area links, ensuring secure and efficient routing of traffic.

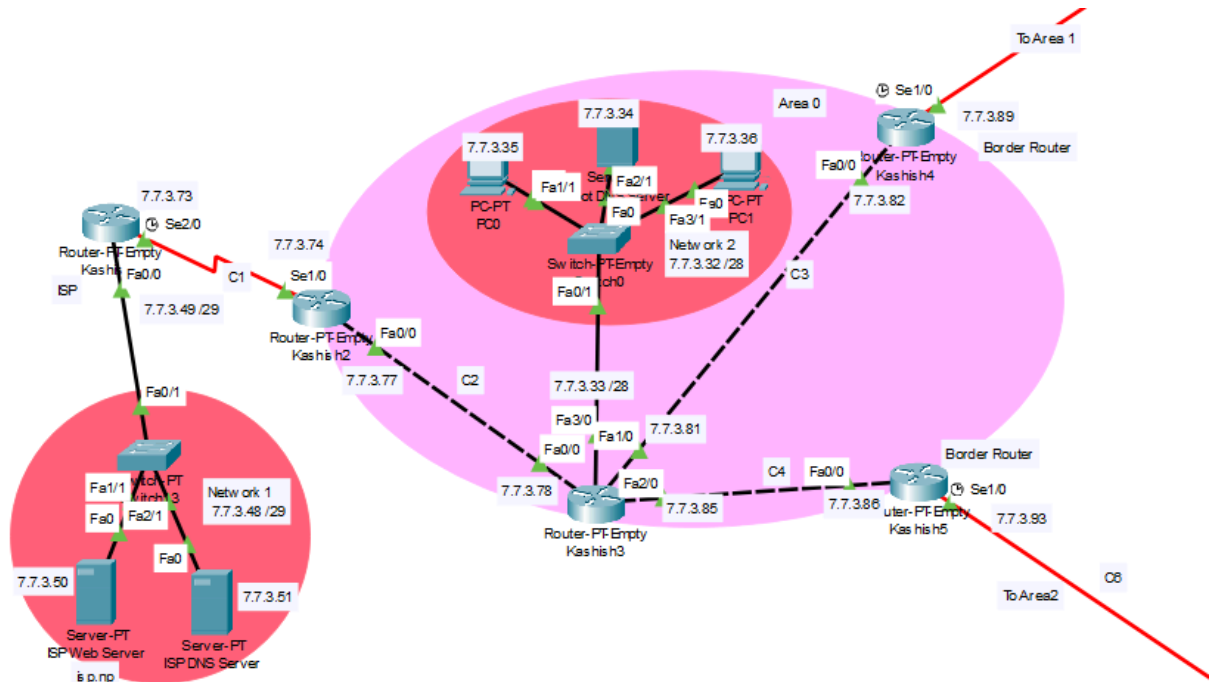
Routing Protocol

The Open Shortest Path First (OSPF) protocol is used as the routing mechanism across the entire network. OSPF ensures optimal routing paths and quick convergence in the event of network topology changes.

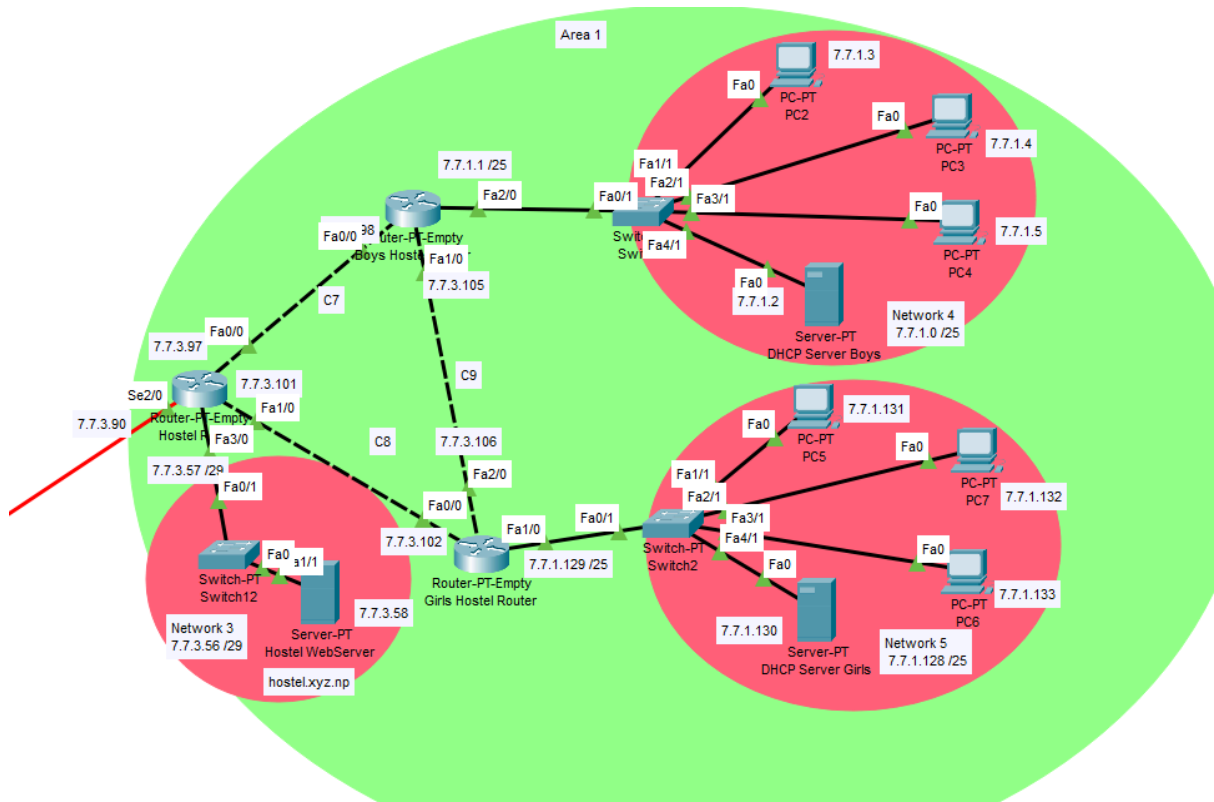
Complete Network Topology



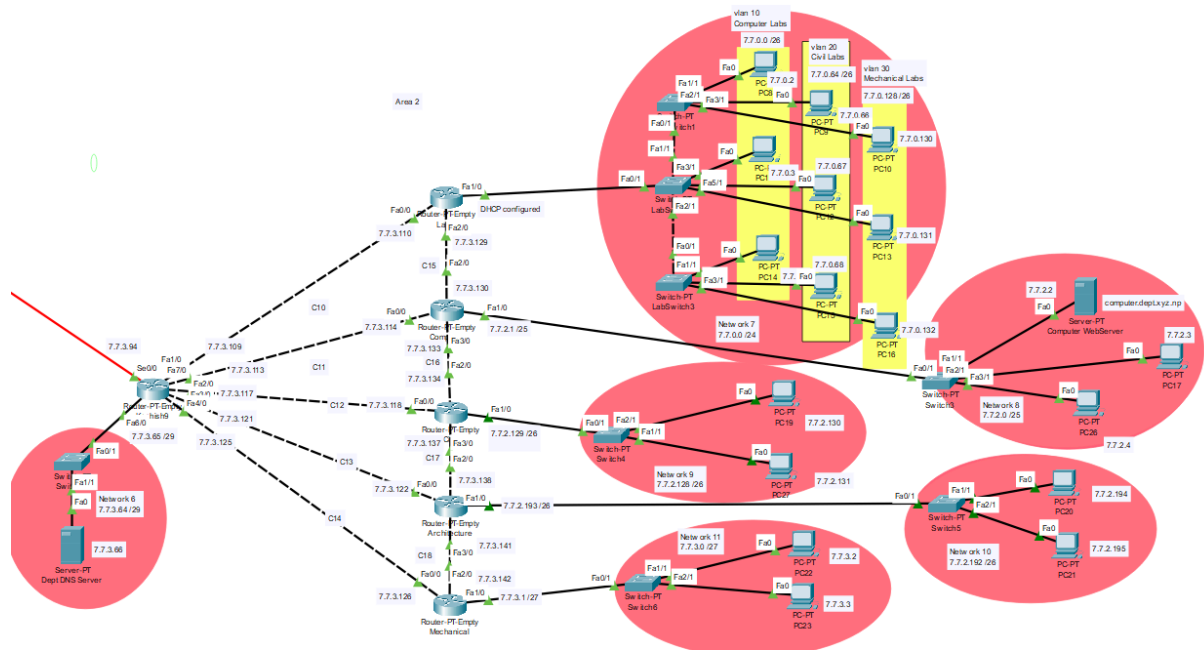
Area 0



Area 1



Area 2



Subnet Division

- Network1 is named N1 and so on.
- Connection1 is named C1 and so on.

Subnet	Hosts Required	Subnet Mask	Total IPs	Network Address	Broadcast Address	Usable IP Range	Wildcard Mask
N7	230	/24	256	7.7.0.0	7.7.0.255	7.7.0.1 - 7.7.0.254	0.0.0.255
N4	120	/25	128	7.7.1.0	7.7.1.127	7.7.1.1 - 7.7.1.126	0.0.0.127
N5	120	/25	128	7.7.1.128	7.7.1.255	7.7.1.129 - 7.7.1.254	0.0.0.127
N8	110	/25	128	7.7.2.0	7.7.2.127	7.7.2.1 - 7.7.2.126	0.0.0.127
N9	60	/26	64	7.7.2.128	7.7.2.191	7.7.2.129 - 7.7.2.190	0.0.0.63
N10	53	/26	64	7.7.2.192	7.7.2.255	7.7.2.193 - 7.7.2.254	0.0.0.63
N11	28	/27	32	7.7.3.0	7.7.3.31	7.7.3.1 - 7.7.3.30	0.0.0.31
N2	12	/28	16	7.7.3.32	7.7.3.47	7.7.3.33 - 7.7.3.46	0.0.0.15
N1	6	/29	8	7.7.3.48	7.7.3.55	7.7.3.49 - 7.7.3.54	0.0.0.7
N3	6	/29	8	7.7.3.56	7.7.3.63	7.7.3.57 - 7.7.3.62	0.0.0.7
N6	6	/29	8	7.7.3.64	7.7.3.71	7.7.3.65 - 7.7.3.70	0.0.0.7
C1	2	/30	4	7.7.3.72	7.7.3.75	7.7.3.73 - 7.7.3.74	0.0.0.3
C2	2	/30	4	7.7.3.76	7.7.3.79	7.7.3.77 - 7.7.3.78	0.0.0.3
C3	2	/30	4	7.7.3.80	7.7.3.83	7.7.3.81 - 7.7.3.82	0.0.0.3
C4	2	/30	4	7.7.3.84	7.7.3.87	7.7.3.85 - 7.7.3.86	0.0.0.3
C5	2	/30	4	7.7.3.88	7.7.3.91	7.7.3.89 - 7.7.3.90	0.0.0.3
C6	2	/30	4	7.7.3.92	7.7.3.95	7.7.3.93 - 7.7.3.94	0.0.0.3
C7	2	/30	4	7.7.3.96	7.7.3.99	7.7.3.97 - 7.7.3.98	0.0.0.3

C8	2	/30	4	7.7.3.100	7.7.3.103	7.7.3.101 - 7.7.3.102	0.0.0.3
C9	2	/30	4	7.7.3.104	7.7.3.107	7.7.3.105 - 7.7.3.106	0.0.0.3
C10	2	/30	4	7.7.3.108	7.7.3.111	7.7.3.109 - 7.7.3.110	0.0.0.3
C11	2	/30	4	7.7.3.112	7.7.3.115	7.7.3.113 - 7.7.3.114	0.0.0.3
C12	2	/30	4	7.7.3.116	7.7.3.119	7.7.3.117 - 7.7.3.118	0.0.0.3
C13	2	/30	4	7.7.3.120	7.7.3.123	7.7.3.121 - 7.7.3.122	0.0.0.3
C14	2	/30	4	7.7.3.124	7.7.3.127	7.7.3.125 - 7.7.3.126	0.0.0.3
C15	2	/30	4	7.7.3.128	7.7.3.131	7.7.3.129 - 7.7.3.130	0.0.0.3
C16	2	/30	4	7.7.3.132	7.7.3.135	7.7.3.133 - 7.7.3.134	0.0.0.3
C17	2	/30	4	7.7.3.136	7.7.3.139	7.7.3.137 - 7.7.3.138	0.0.0.3
C18	2	/30	4	7.7.3.140	7.7.3.143	7.7.3.141 - 7.7.3.142	0.0.0.3

Information about servers

Server Name	Network Name	Network Address	Server IP	Description
ISP Web	N1	7.7.3.48/29	7.7.3.50	Located in the ISP network, hosting its web services.
ISP DNS	N1	7.7.3.48/29	7.7.3.51	Located in the ISP network, managing DNS services for domain resolution.
Backbone DNS	N2	7.7.3.32/28	7.7.3.34	Located in the backbone network, handling internal DNS queries for the campus network.
Hostel Web	N3	7.7.3.56/29	7.7.3.58	Located in the hostel network, providing web services for hostel residents.
DHCP Server Boys	N4	7.7.1.0/25	7.7.1.2	Located in the boys' hostel network, issuing IP addresses to devices in the boys' hostel.
DHCP Server Girls	N5	7.7.1.128/25	7.7.1.130	Located in the girls' hostel network, issuing IP addresses to devices in the girls' hostel.
Department DNS	N6	7.7.3.64/29	7.7.3.66	Located in the department network, managing DNS services for departmental resources.
Computer Web	N8	7.7.2.0/25	7.7.2.2	Located in the computer department, hosting web services for the computer department.

Information about VLAN

VLAN	Network Address	Subnet Mask	Usable IP Range	Broadcast Address
VLAN 1	7.7.0.0	255.255.255.192	7.7.0.1 - 7.7.0.62	7.7.0.63
VLAN 2	7.7.0.64	255.255.255.192	7.7.0.65 - 7.7.0.126	7.7.0.127
VLAN 3	7.7.0.128	255.255.255.192	7.7.0.129 - 7.7.0.190	7.7.0.191

Demonstration of connectivity

1. Inter-VLAN Routing - Successful
2. Interdepartmental Routing - Successful
3. Routing among Boys' and Girls' hostels - Successful
4. Inter-area Routing - Successful
5. Routing to ISP - Successful

Proof of a successful ping from a PC in Mechanical Lab VLAN to a PC in the Boys' Hostel.

```
C:\>PING 7.7.1.4

Pinging 7.7.1.4 with 32 bytes of data:

Reply from 7.7.1.4: bytes=32 time=16ms TTL=121
Reply from 7.7.1.4: bytes=32 time=62ms TTL=121
Reply from 7.7.1.4: bytes=32 time=15ms TTL=121
Reply from 7.7.1.4: bytes=32 time=37ms TTL=121

Ping statistics for 7.7.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 62ms, Average = 32ms
```

Tracert to 1.1.1.1 from a PC in Mechanical Lab VLAN.

```
C:\>tracert 1.1.1.1

Tracing route to 1.1.1.1 over a maximum of 30 hops:

  0  55 ms    10 ms     1 ms     7.7.0.129
  1  0 ms      0 ms      0 ms     7.7.3.109
  2  1 ms      19 ms     128 ms   7.7.3.93
  3  105 ms    13 ms     161 ms   7.7.3.85
  4  126 ms    13 ms     45 ms    7.7.3.77
  5  13 ms     18 ms     11 ms    7.7.3.73
  6  13 ms     *         450 ms   7.7.3.73
  7  *         33 ms     *        Request timed out.
  8  35 ms     *         35 ms    7.7.3.73
  9  *         13 ms     *        Request timed out.
 10  12 ms     *         227 ms   7.7.3.73
```

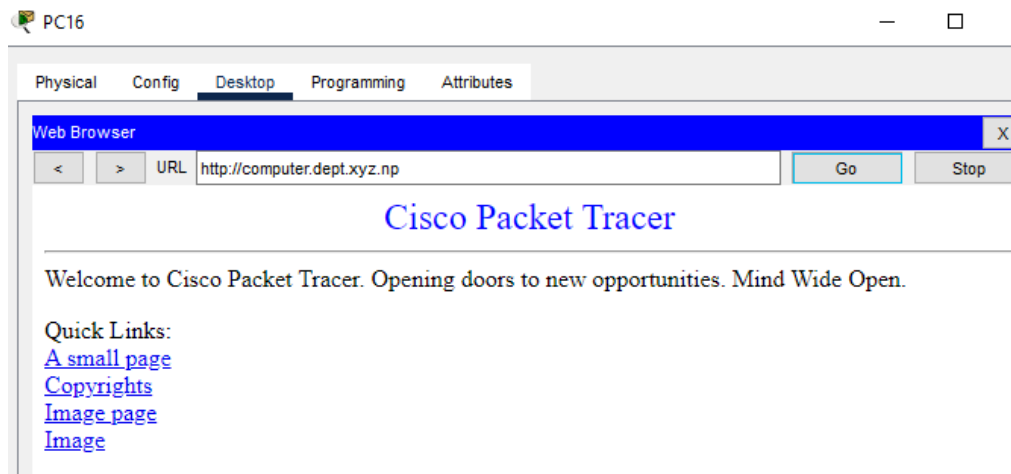
This screenshot shows that any traffic is forwarded to the ISP.

```
ip route 7.7.0.0 255.255.252.0 7.7.3.74
!
ip flow-export version 9
!
!
```

This screenshot shows the running config of the ISP router. Any traffic for the IP range of the hypothetical campus is forwarded to Area 0.

Web and DNS servers

1. The Computer Department's web server being accessed from a PC in the Mechanical Lab VLAN.



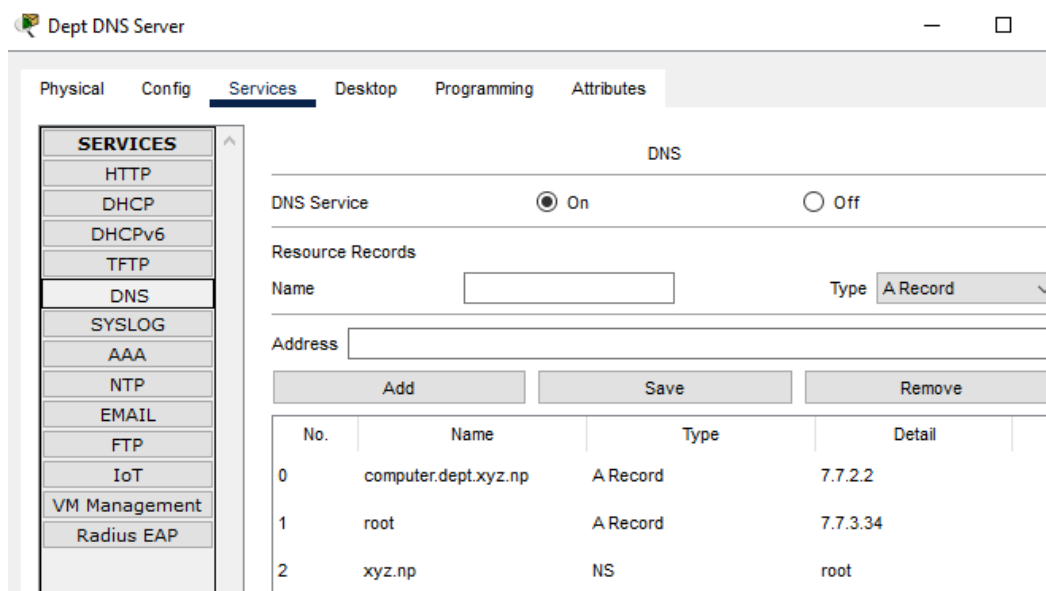
Similarly, the web servers can be accessed from the designated subnets successfully.

3-Level DNS Hierarchy

1. Level 3 - Area 2

The local DNS server in Area 2 handles DNS queries specifically for the subnets and devices within Area 2. If a query cannot be resolved locally, it is forwarded to the Area 0 DNS server. This forwarding is done using NS records.

A screenshot of the Department's DNS server configuration.



The unresolved queries from here are forwarded to Area 0 DNS.

2. Level 2 - Area 0

The intermediate-level DNS server, located in the central backbone (Area 0), handles DNS queries for the entire campus and forwards unresolved queries to the ISP's DNS server. The unresolved queries are forwarded to the ISP.

3. Level 1 - ISP

The top-level DNS server, managed by the ISP, handles external DNS queries and resolves domain names not found within the campus network.

Conclusion

The hypothetical campus network was designed and implemented. The network, divided into three distinct sections: Area 0 (Central Backbone), Area 1 (Hostels), and Area 2 (Departments and Labs) was configured to support communication, dynamic IP allocation, and effective routing mechanisms.

A hierarchical 3-level DNS system was used to ensure that DNS queries were resolved efficiently. The OSPF routing protocol enabled optimal dynamic routing paths maintaining high network performance even if topology changes.

Inter-VLAN, interdepartmental, and inter-area routing were all successfully demonstrated, along with the successful routing of traffic to and from the ISP. The web and DNS servers across the campus and ISP networks performed as expected, with all services being accessible from the designated areas.

In conclusion, this project demonstrated all the knowledge gained throughout the course this semester. It was a fun experience planning and implementing the proposed topology.